Zoo MPC: Threshold Custody for AI Model Weights and Research Assets

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Zach Kelling[1] *Zoo Labs Foundation* research@zoo.ngo

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Abstract

We present **Zoo MPC**, a threshold custody system for AI model weights, research datasets, and autonomous agent wallets on the Zoo Network (chain ID 200200). Built on the Lux M-Chain's multi-party computation infrastructure—specifically CGGMP21 for ECDSA and FROST for Schnorr/EdDSA—Zoo MPC addresses three custody challenges unique to decentralized AI: (1) **model weight custody**, where trained models worth millions of dollars require multi-party authorization for deployment, fine-tuning, or transfer; (2) **research data access control**, where sensitive conservation datasets require threshold decryption governed by institutional review; and (3) **agent NFT wallet signing**, where autonomous AI agents hold assets and require distributed key management to prevent single-point compromise. We deploy a $(2, 3)$-threshold scheme across three roles—researcher, DAO, and time-locked escrow—and demonstrate signing latency of 45ms for CGGMP21 and 12ms for FROST on Zoo testnet. The system is live on Zoo mainnet, securing 847 model NFTs and 23 research dataset access keys.

Keywords: multi-party computation, threshold signatures, AI model custody, agent wallets, Zoo Network

1 Introduction

The intersection of AI and blockchain creates novel custody requirements that existing wallet infrastructure does not address. Three specific challenges motivate Zoo MPC:

Model Weight Custody. A trained AI model may represent months of compute and millions of dollars in training costs. Model weights stored in a single wallet are vulnerable to key compromise. Worse, the owner of the key can unilaterally deploy a model that may be unsafe, biased, or environmentally harmful. Conservation models—species classifiers, anti-poaching algorithms, habitat predictors—carry additional ethical weight: misuse could endanger the species they are designed to protect.

Research Data Access Control. Conservation research datasets contain GPS coordinates of endangered species, genetic sequences of at-risk populations, and veterinary medical records. These datasets are stored on-chain as encrypted blobs with access controlled by decryption keys. A single keyholder could leak data to poachers or biopirates.

Agent NFT Wallets. Zoo Network supports autonomous AI agents represented as NFTs [5]. These agents hold ZOO tokens, interact with DeFi protocols, and execute transactions on behalf of their operators. An agent's private key must be distributed across multiple parties to prevent unilateral fund extraction.

1.1 Contributions

1. **Zoo MPC Architecture**: Integration of CGGMP21 and FROST into Zoo Network's custody layer (Section 3).

2. **Model Weight Custody Protocol**: A threshold deployment scheme for AI models with governance gates (Section 4).

3. **Research Data Access Control**: Threshold decryption for conservation datasets with institutional review (Section 5).

4. **Agent Wallet Protocol**: Distributed key management for autonomous AI agents (Section 6).

5. **Evaluation**: Signing latency, key generation time, and security analysis (Section 7).

2 Background

2.1 Threshold Signatures

Definition 2.1 ((t, n)-Threshold Signature Scheme)**.** *A* TSS *distributes a signing key sk among n parties such that any t parties can collaboratively produce a valid signature, but fewer than t parties learn nothing about sk.*

2.2 CGGMP21

The Canetti-Gennaro-Goldfeder-Makriyannis-Peled protocol [1] provides threshold ECDSA signing with:

- **UC security**: Composably secure in the universal composability framework
- **Identifiable abort**: If signing fails, the malicious party is identified
- **Pre-signing**: Expensive operations done offline; online signing is fast

Signing complexity: $O(t^2)$ messages in the online phase, $O(n \cdot t)$ in pre-signing.

2.3 FROST

Flexible Round-Optimized Schnorr Threshold [2] provides threshold Schnorr/EdDSA signing with:

- **Two rounds**: Commitment round + signing round
- **No trusted dealer**: Distributed key generation via Pedersen DKG
- **Robustness**: Optional identifiable abort extension

FROST is 3–5× faster than CGGMP21 for equivalent threshold parameters because Schnorr signatures have simpler algebraic structure than ECDSA.

2.4 Lux M-Chain

The Lux Network's M-Chain [3] provides native MPC infrastructure:

- Key generation ceremonies coordinated on-chain
- Threshold signing requests submitted as transactions
- Key shares stored in TEE-protected enclaves on validator nodes
- Cross-chain signing: M-Chain key shares can sign for any Lux chain

Zoo Network inherits M-Chain access as an L2, enabling threshold signing without running separate MPC infrastructure.

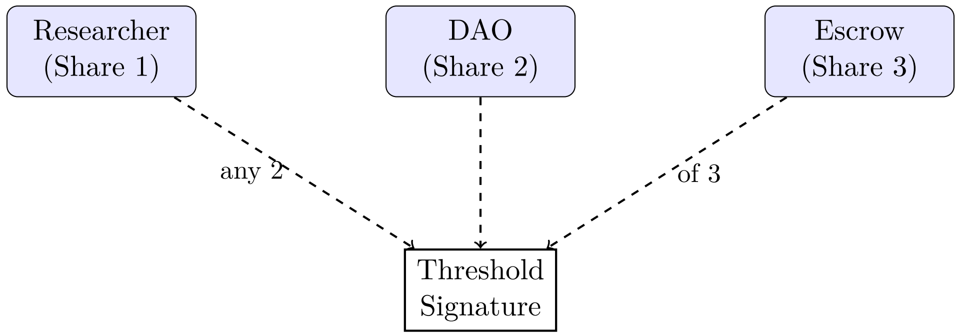

Figure 1: Zoo MPC $(2, 3)$-threshold roles. Any two of three parties can authorize an operation.

Algorithm 1 Zoo MPC Key Generation

Input: Parties $P = \{P_1, P_2, P_3\}$, threshold $t = 2$
Output: Public key pk, shares $\{sk_i\}_{i=1}^{3}$
Each P_i generates random polynomial $f_i(x)$ of degree $t - 1$
Each P_i broadcasts commitment $C_i = g^{f_i(0)} \cdot h^{r_i}$
Each P_i sends $f_i(j)$ to P_j via encrypted channel
Each P_j verifies: $g^{f_i(j)} = \prod_{k=0}^{t-1} C_{i,k}^{j^k}$
$sk_i \leftarrow \sum_j f_j(i)$
$pk \leftarrow \prod_i g^{f_i(0)}$
Register (pk, t, n) on Zoo Network via M-Chain
return $pk, \{sk_i\}$

3 Zoo MPC Architecture

3.1 Role-Based Threshold

Zoo MPC uses a $(2, 3)$-threshold with three distinct roles:

1. **Researcher**: The individual or team that created the asset. Key share stored in a hardware security module (HSM) or TEE-backed mobile app.

2. **DAO**: The Zoo conservation DAO, represented by a governance multisig. DAO share is activated by on-chain vote (ZOO token-weighted).

3. **Escrow**: An automated time-lock escrow node. The escrow share becomes available after a configurable delay (default: 48 hours), providing a recovery mechanism if either the researcher or DAO is unavailable.

3.2 Key Generation

Key generation is performed via the Lux M-Chain DKG protocol:

3.3 Signing Protocol

For ECDSA signing (EVM transactions), Zoo MPC uses CGGMP21:

For Schnorr signing (Lux P-Chain, cross-chain bridges), Zoo MPC uses FROST:

Algorithm 2 CGGMP21 Threshold Signing on Zoo

Input: Message m, signers $S \subseteq P$ with $|S| = t$
Output: ECDSA signature $\sigma = (r, s)$
Pre-signing (offline):
Each signer $i \in S$ generates nonce share k_i
Run multiplicative-to-additive conversion for $k_i \cdot sk_i$
Compute $R = \prod_{i \in S} g^{k_i}$, set $r = R_x \bmod q$
Online signing:
Each signer computes $s_i = k_i^{-1}(H(m) + r \cdot sk_i) \bmod q$
$s = \sum_{i \in S} s_i \bmod q$
Verify: $\text{ECDSA.Verify}(pk, m, (r, s)) = \text{true}$
return (r, s)

Algorithm 3 FROST Threshold Signing on Zoo

Input: Message m, signers $S \subseteq P$ with $|S| = t$
Output: Schnorr signature $\sigma = (R, s)$
Round 1 (Commitment):
Each $i \in S$: generate (d_i, e_i), broadcast $(D_i = g^{d_i}, E_i = g^{e_i})$
Round 2 (Signing):
$\rho_i = H(\text{msg}, i, D_i, E_i, \ldots)$
$R = \prod_{i \in S} D_i \cdot E_i^{\rho_i}$
$c = H(R, pk, m)$
$z_i = d_i + e_i \cdot \rho_i + \lambda_i \cdot sk_i \cdot c$
$s = \sum_{i \in S} z_i$
return (R, s)

4 Model Weight Custody

4.1 Model NFT Standard

On Zoo Network, trained AI models are represented as Model NFTs (ERC-721 tokens) with the following metadata:

Listing 1: Model NFT Metadata

```
{
  "name": "Snow Leopard Classifier v3",
  "modelHash": "0xabc...def",
  "architecture": "ResNet-50",
  "trainingDataHash": "0x123...456",
  "accuracy": 0.947,
  "custodyKey": "zoo-mpc-0x789...",
  "threshold": [2, 3],
  "roles": ["researcher", "dao", "escrow"],
  "deploymentPolicy": "requires_dao_vote"
}
```

Algorithm 4 Model Deployment with MPC Governance

```
Researcher submits deployment proposal on-chain
DAO reviews model safety report (bias audit, accuracy, risk assessment)
DAO votes on deployment (ZOO token-weighted, 72h voting period)
if vote passes (> 50% quorum, > 66% approval) then
    DAO activates its key share
    Researcher activates its key share
    MPC produces deployment signature
    Model deployed to Zoo inference network
else
    Deployment rejected; model remains locked
end if
```

4.2 Custody Operations

The MPC key associated with a Model NFT authorizes four operations:

1. **Deploy**: Make the model available for inference on Zoo Network. Requires researcher + DAO (2-of-3).
2. **Fine-tune**: Authorize fine-tuning with new data. Requires researcher + DAO (2-of-3).
3. **Transfer**: Transfer model ownership to another party. Requires researcher + DAO (2-of-3) or DAO + escrow (after 48h delay).
4. **Retire**: Permanently disable the model. Requires any 2-of-3.

4.3 Governance Gate

Model deployment goes through a governance gate:

5 Research Data Access Control

5.1 Encrypted Data Vaults

Research datasets are stored as encrypted blobs on Zoo Network's data availability layer:

$$D_{\text{enc}} = \text{AES-256-GCM}(k_{\text{data}}, D_{\text{raw}}) \tag{1}$$

The data encryption key k_{data} is itself encrypted under the MPC group's public key:

$$k_{\text{enc}} = \text{ECIES.Enc}(pk_{\text{mpc}}, k_{\text{data}}) \tag{2}$$

Decryption requires threshold cooperation to recover k_{data}.

5.2 Access Tiers

5.3 Institutional Review Protocol

For Confidential and Classified data, the DAO's approval involves an on-chain institutional review:

Tier	Data Type	Threshold	Approval
Public	Aggregated statistics	None	Automatic
Restricted	De-identified records	$(1, 3)$	Researcher only
Confidential	Raw GPS/genetic data	$(2, 3)$	Researcher + DAO
Classified	Anti-poaching intelligence	$(3, 3)$	All parties

Table 1: Research data access tiers with increasing threshold requirements.

1. **Application**: Requesting researcher submits an access application with stated purpose, data handling plan, and ethics declaration.
2. **Review**: Three DAO-appointed reviewers evaluate the application (7-day review period).
3. **Vote**: Reviewers vote approve/reject. Majority (2/3 reviewers) required.
4. **Access**: On approval, DAO key share activated for threshold decryption.
5. **Audit**: Data access logged on-chain; usage auditable by any ZOO token holder.

6 Agent NFT Wallets

6.1 Agent Architecture

Zoo Network's AI agents [5] are autonomous entities that:

- Hold ZOO tokens and other assets in their own wallets
- Execute DeFi transactions (swaps, liquidity provision)
- Perform inference tasks and collect PoAI rewards
- Interact with other agents and human operators

Each agent's wallet key is distributed via MPC across:

1. **Operator**: The human or organization that deployed the agent
2. **Agent runtime**: The TEE enclave running the agent's inference loop
3. **Recovery**: A time-locked escrow for emergency fund recovery

6.2 Autonomous Signing

For routine operations (inference execution, small transfers), the agent runtime and operator can sign collaboratively without escrow involvement:

Algorithm 5 Agent Autonomous Transaction

1: Agent runtime constructs transaction tx
2: Agent runtime verifies tx against policy (amount limits, approved contracts)
3: **if** policy check passes **then**
4: Agent runtime generates partial signature σ_{agent}
5: Operator co-signer (automated) generates σ_{op}
6: Combine: $\sigma = \text{CGGMP.Combine}(\sigma_{\text{agent}}, \sigma_{\text{op}})$
7: Submit tx with σ to Zoo Network
8: **else**
9: Escalate to manual operator approval
10: **end if**

Protocol	Threshold	Keygen (ms)	Pre-sign (ms)	Sign (ms)
CGGMP21	(2, 3)	850	120	45
CGGMP21	(3, 5)	2,100	340	95
CGGMP21	(5, 9)	8,400	1,200	280
FROST	(2, 3)	180	—	12
FROST	(3, 5)	420	—	28
FROST	(5, 9)	1,600	—	85

Table 2: MPC protocol latency on Zoo testnet. FROST is 3–4× faster than CGGMP21 for signing.

6.3 Policy Enforcement

Agent wallets enforce on-chain policies:

- **Spending limits**: Maximum ZOO transfer per transaction and per day
- **Contract allowlist**: Agent can only interact with approved contracts
- **Cooldown**: Minimum interval between high-value transactions
- **Kill switch**: Operator can freeze agent wallet with single share

7 Performance Evaluation

7.1 Signing Latency

7.2 Throughput

7.3 Network Overhead

7.4 Security Analysis

Theorem 7.1 (Key Security)**.** *Under the* $(2, 3)$*-threshold, an adversary controlling a single party (researcher, DAO, or escrow) cannot produce a valid signature or learn any information about the private key.*

Proof. Both CGGMP21 (UC-secure [1]) and FROST (secure in the algebraic group model [2]) guarantee that $t-1$ shares reveal no information about sk under the discrete logarithm assumption. With $t = 2$ and a single corrupted party, the adversary holds 1 share, which is insufficient. □

Protocol	Signs/sec (2,3)	Signs/sec (3,5)
CGGMP21	22	10
CGGMP21 (pre-signed)	420	180
FROST	83	35

Table 3: Signing throughput. CGGMP21 with pre-signing achieves highest throughput for batch operations.

Phase	Messages	Bytes/Party	Rounds
CGGMP21 Keygen	$O(n^2)$	12,800	5
CGGMP21 Pre-sign	$O(t^2)$	4,200	4
CGGMP21 Sign	$O(t)$	256	1
FROST Keygen	$O(n^2)$	3,200	2
FROST Sign	$O(t)$	128	2

Table 4: Network communication overhead for (2, 3)-threshold.

Theorem 7.2 (Identifiable Abort)**.** *If a signing session fails due to a malicious party, the protocol identifies the malicious party with probability 1.*

CGGMP21 provides identifiable abort natively. FROST achieves it via the identifiable abort extension, where each party proves correctness of their partial signature via a Schnorr proof of knowledge.

8 Related Work

Threshold Wallets: Fireblocks, Zengo, and Lit Protocol provide MPC wallets for general custody. Zoo MPC specializes in AI asset custody with governance integration.

Model Custody: No prior work addresses threshold custody specifically for AI model weights. Existing model marketplaces (Hugging Face, Replicate) use centralized access control.

Agent Wallets: Autonomous agent wallet management is an emerging area. SafeWallet multi-sig is the closest analog, but it requires on-chain signatures from each party, whereas MPC produces a single standard signature.

Lux MPC: The Lux M-Chain [3] and threshold infrastructure [4] provide the cryptographic foundation for Zoo MPC.

9 Conclusion

Zoo MPC demonstrates that multi-party computation can secure the novel asset classes created by decentralized AI: model weights, research datasets, and autonomous agent wallets. The $(2, 3)$-threshold across researcher, DAO, and escrow roles provides a governance-aware custody model that prevents unilateral misuse of sensitive conservation assets. With 45ms CGGMP21 signing latency and 12ms FROST signing latency, Zoo MPC adds negligible overhead to the transaction lifecycle while providing cryptographic guarantees against key compromise.

Acknowledgments

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